Do Now · retrieval

Graphs: directed and undirected

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which structure is LIFO?

Q2

Which structure is FIFO?

Q3

In an array-based linked list, what does the next pointer store?

Q4

In an undirected graph, if A is connected to B, is B also connected to A?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Create — CODE:
int[,] graph = new int[3,3]; — Three vertices (A, B and C) produce a 3 by 3 matrix: one row and one column per vertex. The four-vertex 4×4 example comes later in You Do.

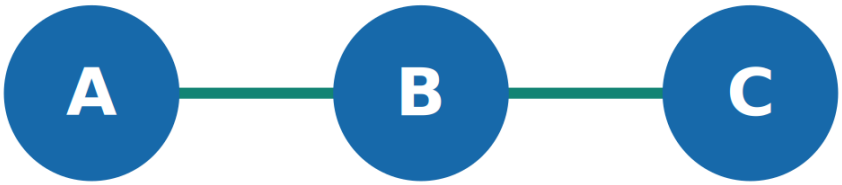
Core idea 2

Add undirected edge — CODE:
graph[a,b] = 1;
graph[b,a] = 1; — The matrix is symmetrical for an undirected connection.

Core idea 3

Add directed edge — CODE:
graph[a,b] = 1; — Only A→B is stored.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Remove and inspect — CODE:
graph[a,b] = 0;

for (int v = 0; v < graph.GetLength(1); v++)
{
if (graph[a,v] == 1)
Console.WriteLine(v);
} — Clearing removes an edge; scanning a row finds neighbours.

Vocabulary · use the correct technical terms

graph — Vertices connected by edges.

vertex — A point/node in a graph.

edge — A connection between vertices.

directed — Edges have a direction.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Build A—B—C

Follow the worked method

1. Make matrix — CODE:
int[,] graph = new int[3,3];

2. B—C — CODE:
graph[1,2] = 1;
graph[2,1] = 1;

3. Read neighbours of B — CODE:
row 1 = 1 0 1
B connects to A and C

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Fill the same code
CODE:
graph[0,2] = ?;
graph[2,0] = ?;

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

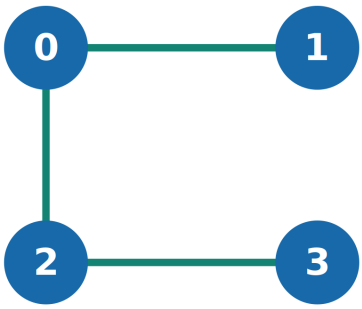
You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Create an undirected graph
Create a 4×4 matrix and add 0—1, 0—2 and 2—3.

Question 1 · original source diagram



Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Show direction matters

Create a 2×2 directed matrix with 0→1 only, then output both directional entries.

Question 2 · original source diagram



Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

An undirected edge normally changes two entries in an adjacency matrix.

Use key vocabulary from memory

1. Explain graph in your own words.

2. Explain vertex in your own words.

3. Explain edge in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.